

NELSON, New Zealand

WMO: 935460

Lat: 41.3009S

Lon: 173.2160E

Elev: 6

StdP: 101.25

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.3	1.4	-3.0	2.9	4.5	-1.7	3.3	4.3	12.4	12.8	10.8	12.6	1.2	200	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.6	24.4	18.4	23.3	18.2	22.5	17.8	20.1	22.7	19.5	22.0	18.8	21.3	6.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	21.5	18.5	13.3	21.0	17.8	12.7	20.5	57.9	22.8	55.6	22.0	53.4	21.4	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.2	8.9	DB	-1.5	27.3	0.8	1.6	-2.0	28.4	-2.5	29.4	-3.0	30.3	-3.5	31.4	(4)
(5)			WB	-2.2	21.3	0.9	0.9	-2.8	22.0	-3.3	22.5	-3.8	23.0	-4.4	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	18.1	18.3	16.7	14.0	11.5	8.7	8.0	9.0	10.7	12.6	14.4	16.7	(6)	
	DBStd	4.11	2.05	1.95	1.95	2.00	2.38	2.37	2.08	1.94	1.96	1.98	2.18	2.02	(7)	
	HDD10.0	197	0	0	0	1	12	54	69	42	15	3	1	0	(8)	
	HDD18.3	1933	28	22	58	130	212	289	322	289	229	177	119	57	(9)	
	CDD10.0	1364	252	232	207	121	58	14	6	11	36	84	133	209	(10)	
	CDD18.3	57	22	21	6	0	0	0	0	0	0	0	1	7	(11)	
	CDH23.3	103	47	36	6	0	0	0	0	0	0	0	2	13	(12)	
	CDH26.7	7	5	1	0	0	0	0	0	0	0	0	0	1	(13)	
(14) Wind	WSAvg	3.3	4.0	3.5	3.3	2.8	2.6	2.4	2.4	2.6	3.2	4.0	4.2	4.2	(14)	
(15) Precipitation	PrecAvg	1132	79	66	82	100	100	116	89	106	97	107	92	100	(15)	
	PrecMax	1603	163	179	181	227	250	203	261	246	184	265	201	401	(16)	
	PrecMin	797	23	6	18	9	10	50	10	38	21	22	16	17	(17)	
	PrecStd	210	41	43	43	60	51	43	51	43	42	61	49	77	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	26.3	24.0	21.2	18.9	16.0	14.7	16.0	17.9	20.5	22.8	24.8	(19)	
		MCWB	17.8	19.7	17.4	15.9	14.5	12.4	11.1	11.1	12.6	13.8	15.7	17.3	(20)	
	2%	DB	24.4	24.3	22.6	19.8	17.6	14.7	13.8	14.6	16.6	18.6	21.1	22.8	(21)	
		MCWB	18.5	18.9	17.5	15.5	14.3	12.0	10.8	10.9	12.4	13.7	15.5	17.3	(22)	
	5%	DB	23.3	23.2	21.7	18.9	16.6	14.0	13.0	13.6	15.5	17.6	19.8	21.7	(23)	
		MCWB	18.3	18.6	17.2	15.1	13.8	11.8	10.3	10.6	12.1	13.5	14.9	17.1	(24)	
	10%	DB	22.3	22.4	20.9	18.0	15.7	13.1	12.1	12.8	14.6	16.6	18.7	20.7	(25)	
		MCWB	17.9	18.1	16.8	14.7	13.3	10.9	9.5	10.3	11.7	13.1	14.5	16.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	21.5	19.8	18.1	16.6	14.2	13.2	13.1	14.5	16.1	17.9	20.0	(27)	
		MCDB	24.3	24.0	22.2	19.2	17.4	14.8	13.9	14.0	16.0	18.1	21.0	22.4	(28)	
	2%	WB	20.1	20.3	19.0	17.1	15.6	13.3	12.0	12.2	13.7	15.2	17.0	18.9	(29)	
		MCDB	22.7	22.7	21.2	18.6	16.6	14.1	12.9	13.3	15.0	17.1	19.5	21.2	(30)	
	5%	WB	19.5	19.6	18.3	16.3	14.6	12.4	11.0	11.4	13.0	14.5	16.1	18.1	(31)	
		MCDB	22.0	22.2	20.7	17.8	16.0	13.4	12.2	12.8	14.5	16.4	18.3	20.3	(32)	
	10%	WB	18.7	18.8	17.6	15.5	13.6	11.2	9.9	10.7	12.4	13.9	15.3	17.5	(33)	
		MCDB	21.2	21.6	20.1	17.3	15.2	12.6	11.5	12.3	14.1	15.8	17.7	19.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.6	7.9	7.8	7.8	7.9	8.1	7.8	7.6	7.6	7.7	7.1	(35)	
		MCDBR	8.6	8.4	8.5	8.4	7.9	7.5	8.3	8.3	8.4	8.6	8.6	8.2	(36)	
	5% WB	MCWBR	5.4	5.4	5.3	5.5	5.3	5.4	5.7	5.7	5.8	5.7	5.5	5.3	(37)	
		MCWBR	6.8	7.1	7.1	6.5	6.3	5.6	6.7	6.7	6.9	7.1	6.9	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.337	0.333	0.316	0.306	0.293	0.289	0.284	0.294	0.319	0.324	0.326	0.333	(40)	
		taud	2.576	2.600	2.650	2.661	2.658	2.638	2.655	2.649	2.548	2.560	2.569	2.592	(41)	
	Ebn at Noon		991	970	945	889	835	803	837	889	925	968	996	1000	(42)	
		Edh at Noon	103	96	84	71	61	57	59	70	91	99	103	103	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.60	5.90	4.59	3.06	2.08	1.59	1.83	2.60	3.72	4.98	6.14	6.36	(44)	
	RadStd		0.50	0.51	0.24	0.29	0.12	0.12	0.13	0.17	0.28	0.38	0.53	0.51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.32	N/A	N/A	+0.60	N/A	N/A	N/A	-88	+70	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page